

National Taipei University of Technology
Computer Science and Information Engineering

Spatial Databases

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Semester Group Project Report

LAQP for Spatial Data

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Abstract

Approximate Query Processing (AQP) enables fast responses to aggregation queries over massive datasets, addressing the challenge that exact query execution often becomes prohibitively slow in big data environments. This project reproduces and extends LAQP (Learning-based Approximate Query Processing), a method that trains an error prediction model from historical queries to refine sampling-based estimates, thereby improving approximation accuracy over traditional AQP techniques.

We implement LAQP in a spatial database context and evaluate it against two baselines—SAQP (pure sampling-based AQP) and AQP++ (pre-aggregation AQP based on range similarity)—using two real-world datasets: the POWER household electric power consumption dataset and the Uber Pickups in New York City dataset. Experimental results show that LAQP consistently achieves lower Average Relative Error (ARE) than the baselines on both datasets, demonstrating superior estimation accuracy. This improvement, however, comes at the cost of increased query processing time due to error model inference.

To demonstrate practical applicability, we further develop an interactive Streamlit-based demo for spatial range queries, such as bounding-box aggregations on pickup locations. The demo highlights LAQP’s usefulness in real-world spatial analytics scenarios, including urban mobility analysis and location-based services. Overall, the results validate learning-based AQP as an effective approach for spatial databases, achieving a favorable balance between accuracy and interactivity while identifying opportunities for future runtime optimization.

1. Introduction

Approximate Query Processing (AQP) addresses the challenge of efficiently querying large-scale datasets where exact query execution is prohibitively slow or computationally expensive. While traditional AQP techniques trade accuracy for performance, recent advances leverage machine learning to improve estimation quality and robustness. This project is motivated by the need for accurate, interactive query processing in data-intensive and spatial data applications, where both response time and estimation reliability are critical.

The objective of this project is to implement, reproduce, and evaluate Learning-based Approximate Query Processing (LAQP), a machine-learning-driven AQP framework that enhances approximate query results through learned error prediction. Although the original LAQP method targets general large-scale data querying, this work extends and applies LAQP to a spatial database context, enabling approximate yet reliable spatial query processing. Specifically, this project reproduces the LAQP framework and benchmarks it against two baseline approaches, Sampling-based AQP (SAQP) and AQP++, under both non-spatial and spatial workloads. Experiments are conducted on two real-world datasets: the POWER household electric power consumption dataset and the New York City Uber Pickups dataset. To demonstrate practical applicability, the project also includes an interactive spatial query demo system, supporting bounding-box queries over large-scale spatial data and visualizing approximate results with predicted error bounds. The main contributions of this project are threefold:

- Reproduction and empirical evaluation of LAQP, providing an independent validation of its effectiveness.
- Extension of LAQP to spatial query processing, demonstrating its applicability to spatial databases and geo-spatial analytics.
- Development of an interactive spatial AQP demo, illustrating how learning-based AQP can support real-time exploratory data analysis on large spatial datasets.

Through comprehensive experiments and system implementation, this project demonstrates that learning-based AQP methods such as LAQP can significantly improve approximation accuracy while maintaining low query latency, making them suitable for practical, interactive, and spatial data-intensive applications.

2. Literature review and related works

AQP has been extensively studied as a means to reduce query latency over large-scale datasets by trading accuracy for efficiency. Traditional sampling-based AQP methods estimate query results directly from samples and provide statistical error bounds, but their accuracy often degrades when sample sizes are small or data distributions are skewed. To address these limitations, recent work has explored hybrid approaches that incorporate precomputation and learning-based techniques.

AQP++[1] represents an important step beyond classical sampling-based AQP by bridging sampling with aggregate precomputation. Instead of estimating query results solely from samples, AQP++ computes the difference between a user query and a precomputed aggregate and then corrects the approximation using the exact precomputed value. By exploiting correlations between user queries and precomputed aggregates, AQP++ significantly improves estimation accuracy compared to traditional AQP, while incurring only modest preprocessing overhead. However, its effectiveness strongly depends on the availability of appropriate precomputed aggregates and its ability to identify queries that are close in range or structure. As a result, AQP++ may struggle when query workloads exhibit high diversity or when range similarity does not imply error similarity.

LAQP[2] further advances this line of research by introducing a learning-based mechanism to guide error correction. Instead of relying on range similarity, LAQP trains a regression model on historical query logs to predict the sampling error of incoming queries. Based on this predicted error, LAQP selects a precomputed query with a similar error profile and applies an error-based correction strategy. This approach allows LAQP to achieve higher accuracy than both traditional AQP and AQP++, particularly under small sample sizes, while still preserving statistical error guarantees. The key insight of LAQP is that error similarity is often more informative than query range similarity, especially in complex or non-uniform data distributions.

Compared to AQP++, LAQP offers greater robustness to diverse query workloads by decoupling query similarity from spatial or range-based proximity. While AQP++ relies on structural similarity between queries and aggregates, LAQP leverages learned patterns in historical errors, making it more adaptable to changing query distributions. However, this improvement comes at the cost of additional model training and dependency on the quality and representativeness of the query log.

3. Problem statement

Datasets usually contain millions to billions of records, and exact query processing requires scanning large portions of the data or accessing complex spatial indexes. As a result, exact query execution is often too slow to support interactive or real-time exploration, where users expect responses within hundreds of milliseconds. To meet latency requirements, systems commonly rely on AQP techniques, such as sampling or pre-aggregated summaries. While these methods significantly reduce query latency, they often suffer from high estimation errors, especially when data distributions are highly skewed or spatially clustered.

The core problem addressed in this project is: how can we efficiently process interactive spatial aggregation queries over large-scale datasets while achieving low estimation error and acceptable query latency, particularly under non-uniform and clustered spatial data distributions? Traditional sampling-based AQP methods incur large variance when sample sizes are small, while pre-aggregation-based methods may fail to capture fine-grained spatial patterns or adapt to diverse query workloads. Thus, there is a fundamental trade-off between accuracy and efficiency that existing approaches struggle to balance.

To achieve better performance, the problem explored in this project is to develop and evaluate a learning-based AQP approach that leverages historical query information and machine learning models to predict sampling error and apply error correction, thereby outperforming traditional sampling-based and pre-aggregation-based AQP methods in terms of accuracy, while maintaining interactive query latency.

4. Proposed models (approaches)

To address the challenge of efficient and accurate approximate query processing over large-scale spatial datasets, this project adopts a learning-based AQP framework and compares it against two representative baseline approaches. The core design focuses on balancing query latency, estimation accuracy, and robustness to skewed spatial distributions.

LAQP: The primary approach used in this project is LAQP (Learning-based Approximate Query Processing). LAQP enhances traditional sampling-based AQP by incorporating a machine learning-based error prediction model to correct estimation bias and variance.

SAQP: SAQP represents the simplest form of approximate query processing, the method uses random sampling to estimate query results, no error correction or learning component is applied.

AQP++: AQP++ improves upon traditional AQP by combining sampling-based estimation with precomputed aggregates, the dataset is partitioned into ranges, aggregates are precomputed for these ranges. For a new query, AQP++ identifies the most range-similar precomputed aggregate and uses it to correct the estimate.

The system is modularly designed to support experimentation and evaluation:

- Query Generation & Execution: Constructs spatial and temporal queries.
- Error Estimation Module: Trains and applies regression models for error prediction.
- Optimization Module: Selects optimal correction strategies.
- Evaluation Module: Computes metrics such as Absolute Relative Error (ARE) and Mean Squared Error (MSE).
- Visualization Module: Plots error distributions and performance comparisons.

A Streamlit-based demo application (demo.py) is developed to demonstrate the practical usability of the proposed approach. Key features include interactive map-based bounding-box selection, time-range filtering, LAQP query results and the matching points that showed on the map according to the query.

5. Experiments

The primary objectives of the experiments include evaluate the accuracy of LAQP compared with traditional AQP approaches (SAQP and AQP++), verify the effectiveness of learning-based

error correction for both spatial and temporal aggregation queries and demonstrate the practicality of LAQP in interactive query scenarios using real-world datasets.

To achieve these objectives, the experiments are designed with multiple datasets and query types. The same query workloads are executed using SAQP, AQP++, and LAQP. We compute exact query results as ground truth and evaluate the approximate results using unified error metrics, such as ARE. The experiments are first conducted on temporal data, as referenced in the original LAQP paper, and then extended to spatial data.

Data

The dataset used are the [POWER dataset \(from UCI Machine Learning Repository\)](#) and the [Uber Pickups in New York City dataset \(Kaggle\)](#). The POWER dataset includes time-series and numeric attributes, with Global_active_power as the aggregation column and SUM as the query type. The Uber dataset includes pickup locations (latitude, longitude) and datetime information (from April to September), with COUNT as the query type.

Dataset loaded: 2049280 rows				
First 5 data:				
	timestamp	Global_active_power	Global_reactive_power	Voltage \
0	0.0	4.216	0.418	234.84
1	60.0	5.360	0.436	233.63
2	120.0	5.374	0.498	233.29
3	180.0	5.388	0.502	233.74
4	240.0	3.666	0.528	235.68
	Global_intensity	Sub_metering_1	Sub_metering_2	Sub_metering_3
0	18.4	0.0	1.0	17.0
1	23.0	0.0	1.0	16.0
2	23.0	0.0	2.0	17.0
3	23.0	0.0	1.0	17.0
4	15.8	0.0	1.0	17.0

Figure 1: First five data in POWER dataset. The original timestamp column is Date/Time, we changed it into timestamp format to evaluate.

Dataset loaded: 4534327 rows				
	timestamp	Lat	Lon	Base
0	660.0	40.7690	-73.9549	B02512
1	1020.0	40.7267	-74.0345	B02512
2	1260.0	40.7316	-73.9873	B02512
3	1680.0	40.7588	-73.9776	B02512
4	1980.0	40.7594	-73.9722	B02512

Figure 2: First five data in Uber dataset. The Date/Time column also changed to timestamp, column Base will not be used.

Environment

Our experiments are implemented in a Python 3.10 environment, utilizing NumPy and Pandas for data analysis and scikit-learn for the error model. The demo establishes a website using Streamlit, displaying map information related to the Uber dataset.

Evaluation Metrics

We use Average Relative Error (ARE) and Mean Squared Error (MSE) to evaluate the performance between SAQP, AQP++ and LAQP. Time during query search and estimate is also compared.

$$\text{Average Relative Error: } ARE = \frac{1}{|Q|} \sum_{q_i \in Q} \frac{|\hat{r}_i - r_i|}{r_i}$$
$$\text{Mean Squared Error: } MSE = \frac{1}{|Q|} \sum_{q_i \in Q} |\hat{r}_i - r_i|^2$$

Figure 3: Metrics formula.

Diversification

In LAQP, this feature prevents the system from making major prediction errors on specific queries. By combining different prediction paths, LAQP effectively smooths out extreme error peaks and reduces the overall maximum error, resulting in more stable and accurate outcomes.

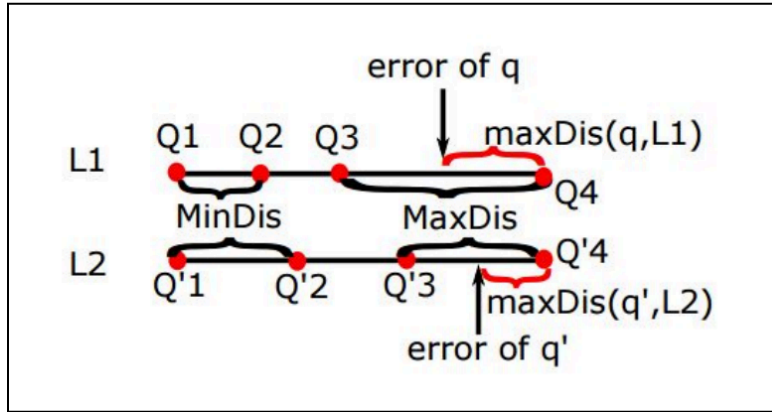


Figure 4: Diversification in LAQP.

Optimization

LAQP is not solely based on an error model but also leverages range similarity for improved performance.

Originally, LAQP relied primarily on the model's predicted error to identify reference points. However, if the model is not well-trained—for instance, due to shallow parameters—the predictions may be inaccurate. To address this limitation, the optimization feature considers both 'error similarity' and 'query range similarity' simultaneously.

$$\begin{aligned} \text{opt}(q, \alpha, \beta) &= \arg \min_{Q_i \in \text{Train}} \alpha \cdot \text{EDis}(q, Q_i) + \beta \cdot \text{RDis}(q, Q_i) \\ \text{EDis}(Q_i, Q_j) &= |\text{error}_{Q_i} - \text{error}_{Q_j}|^2 \\ \text{RDis}(Q_i, Q_j) &= \frac{\sum_{x \in d} (l_{i,x} - l_{j,x})^2 + (r_{i,x} - r_{j,x})^2}{2d} \\ \text{s.t. } \alpha + \beta &= 1, 0 \leq \alpha \leq 1, 0 \leq \beta \leq 1. \end{aligned}$$

Figure 5: Optimization in LAQP. Find the best alpha to get the best performance between error-similar and range-similar.

Non-spatial Process

The POWER dataset was selected to maintain consistency with the LAQP paper. We began by extracting 2000 random samples from the raw data to serve as the base sample for the AQP engine.

```
# Sample size = 2000 as paper used
sample_size = 2000
sample = data.sample(n=sample_size, random_state=42).copy()
print(f"Sample created: {sample.shape[0]} rows")
print(sample.head())
```

Sample created: 2000 rows

	timestamp	Global_active_power	Global_reactive_power	Voltage	\
1030580	61834800.0	1.502	0.074	240.17	
1815	108900.0	0.374	0.264	245.50	
1295977	77758620.0	0.620	0.300	239.85	
206669	12400140.0	0.280	0.200	235.72	
1048893	62933580.0	1.372	0.054	243.95	

	Global_intensity	Sub_metering_1	Sub_metering_2	Sub_metering_3
1030580	6.4	0.0	0.0	18.0
1815	1.8	0.0	2.0	0.0
1295977	3.0	0.0	1.0	1.0
206669	1.4	0.0	0.0	0.0
1048893	5.6	0.0	0.0	18.0

Figure 6: Generate 2000 samples.

To train and evaluate the error model, we constructed three distinct query logs, knowing that in high-dimensional datasets, purely random query generation often results in empty sets (zero records). To ensure that queries intersect with dense data regions, we restricted the query boundaries to 25% (1/4) of the total data range for each dimension. This ensures meaningful results for error analysis:

- Training Set: 800 diverse queries selected from an initial pool of 2000 random queries.
- Testing Set: 100 queries used for model validation during the training phase.
- Evaluation Set: 100 queries reserved for the final performance assessment.

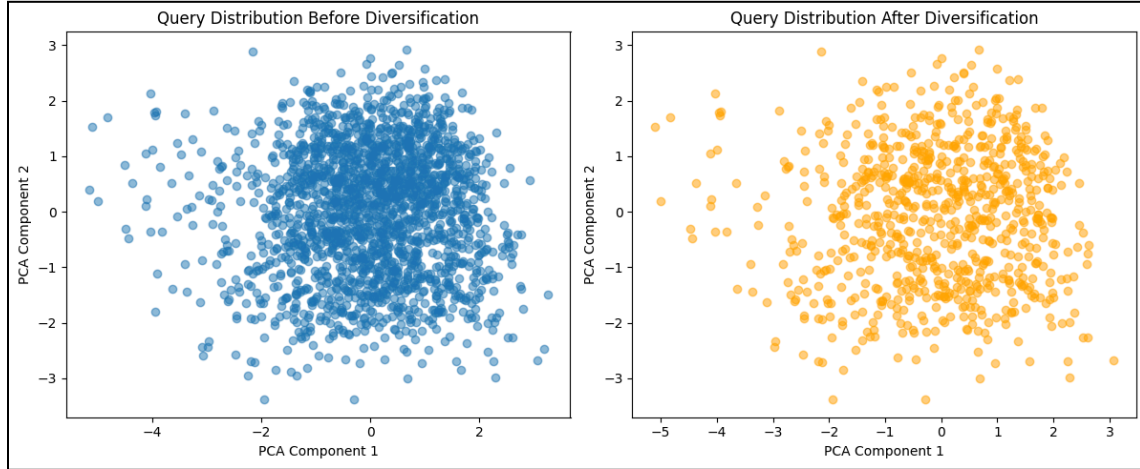


Figure 7: Query distribution before and after diversification.

We implemented the error model using a Regression Forest approach. Adhering to the hyperparameters suggested in the LAQP paper, the forest was configured with a maximum depth (max_depth) of 3 to prevent overfitting while capturing the underlying error distribution.

The final stage of the process involves two key steps:

- Alpha Optimization: We performed an optimization routine to determine the optimal alpha parameter, which balances the influence of the learned error prediction against the range-similar method. Figure 8 shows the process of finding the best alpha.
- Performance Assessment: Using the 100 evaluation queries, we measured the system's performance across several key metrics: ARE, MSE, Min/Max Relative Error and Query Latency (Time).

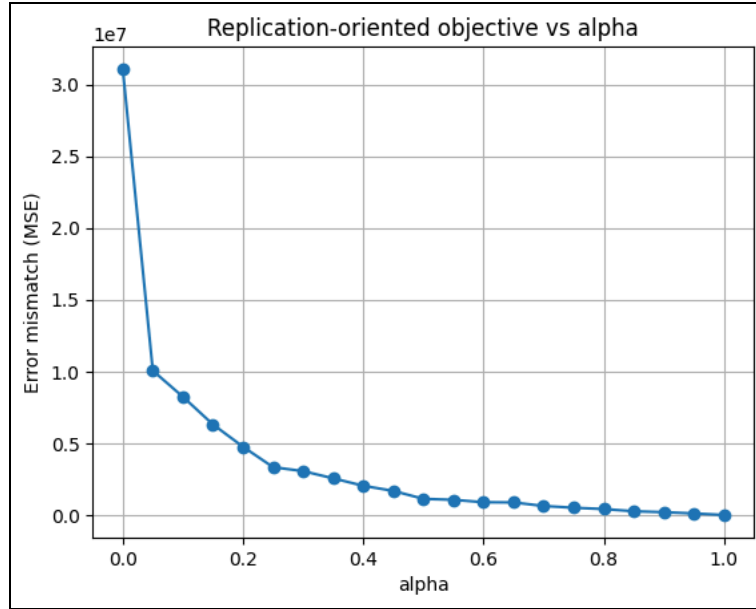


Figure 8: Best alpha curve.

Spatial Process

We use the Uber dataset and extract 1% random samples from the raw data.

```
sample_size = int(0.01 * full_data_size) # 1% for sample
sample = data.sample(n=sample_size, random_state=42).copy()
print(f"Sample created: {sample.shape[0]} rows")
print(sample.head())
```

```
Sample created: 45343 rows
   timestamp    Lat    Lon    Base
255777    850500.0  40.7588 -73.9726 B02617
2591794   7138080.0  40.8451 -73.9418 B02617
1384006  12956280.0  40.7399 -73.9823 B02764
530592   2392200.0  40.6449 -73.7824 B02682
3679163  14676000.0  40.7636 -73.9798 B02598
```

Figure 9: Generate 4000 samples for Uber dataset.

We constructed three distinct query logs similar to the non-spatial process but adjusted the query log sizes due to the larger dataset:

- Training Set: 1000 diverse queries selected from an initial pool of 4000 random queries.
- Testing Set: 250 queries used for model validation during the training phase.
- Evaluation Set: 250 queries reserved for the final performance assessment.

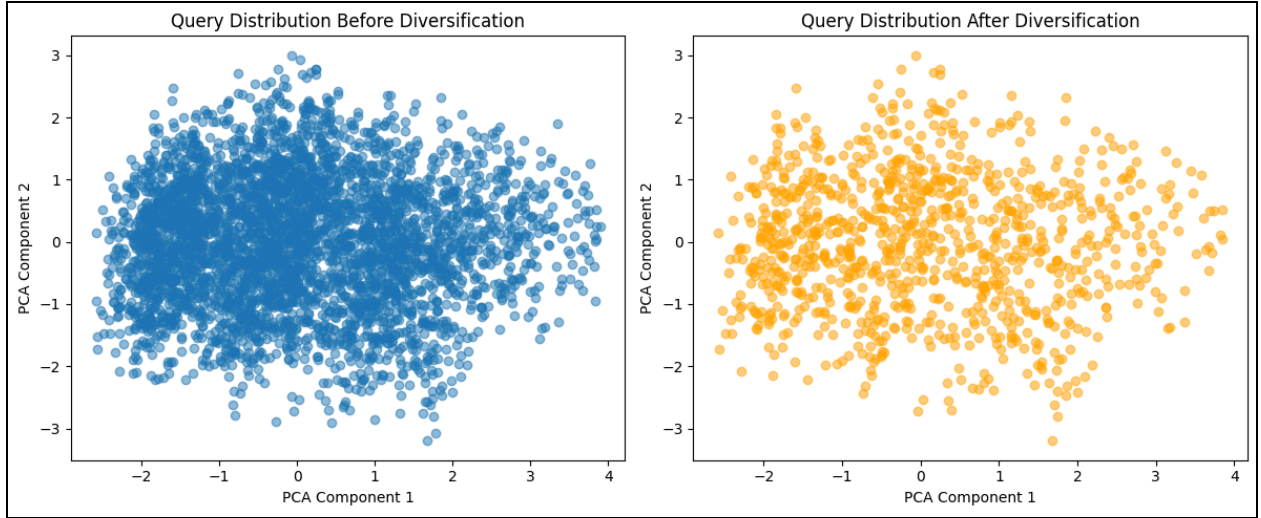


Figure 10: Query distribution before and after diversification.

We used the same method for training the error model but fine-tuned the `max_depth` to 10 for better performance. We also implemented optimization and performance assessment to compare all three methods.

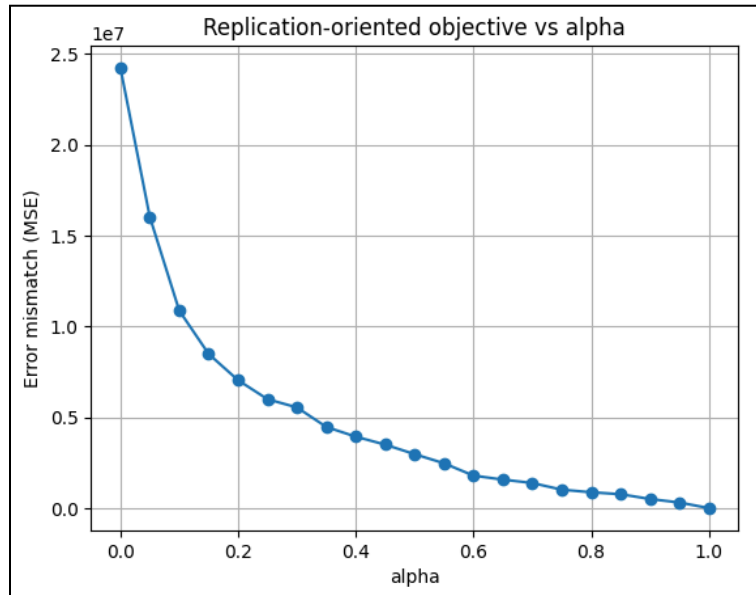


Figure 11: Best alpha curve for Uber dataset.

Experiments Results

For the POWER dataset, the evaluation plots are shown below. LAQP achieves the lowest ARE and MSE, verifying its superior performance. While AQP++ achieves the minimum relative error, it exhibits the highest maximum relative error, which is significantly worse than

LAQP's. LAQP incurs the highest average query time compared to the other two methods, as it involves model prediction.

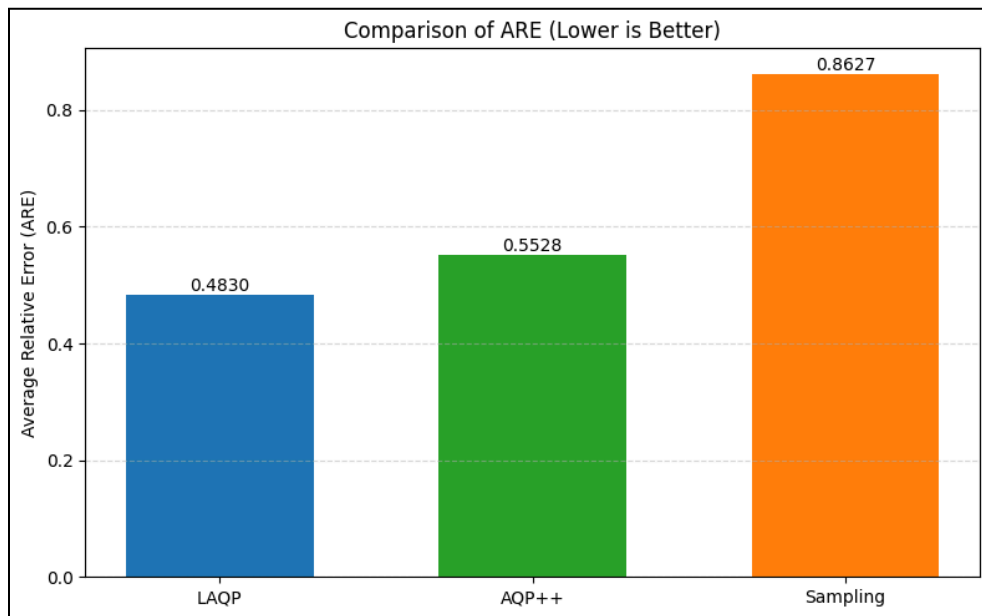


Figure 12: POWER dataset's ARE.

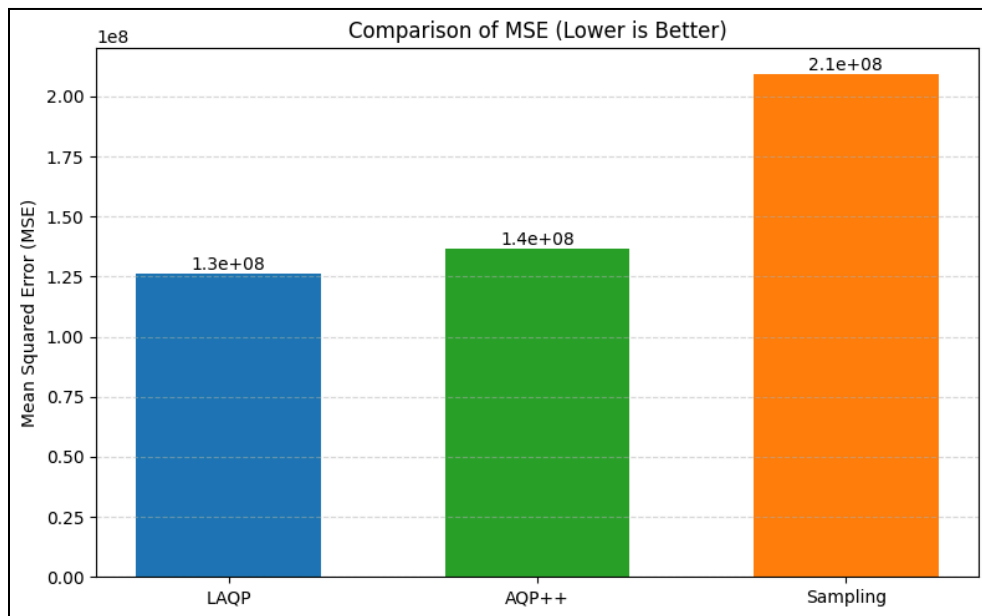


Figure 13: POWER dataset's MSE.

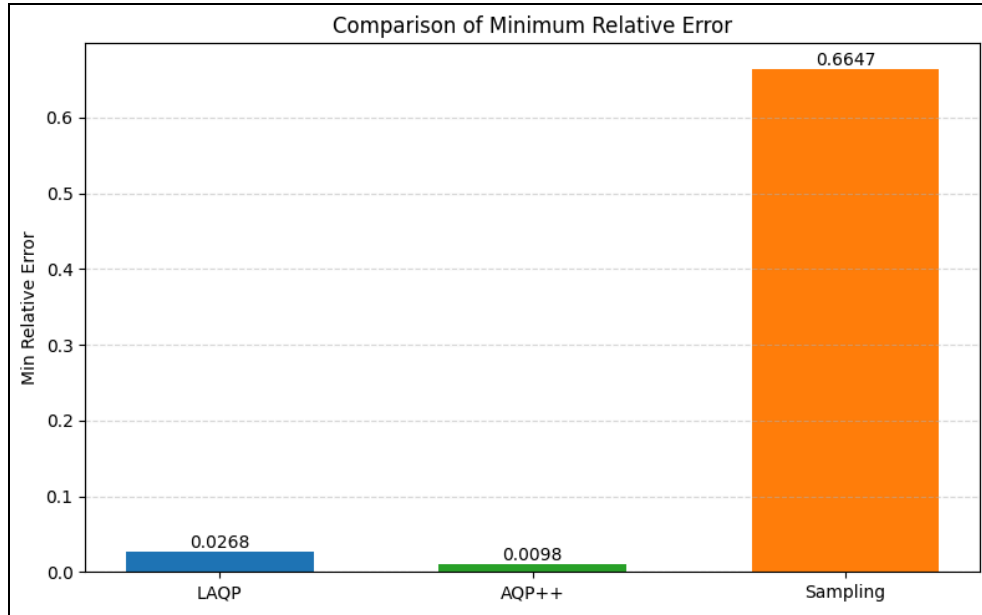


Figure 14: POWER dataset's minimum relative error.

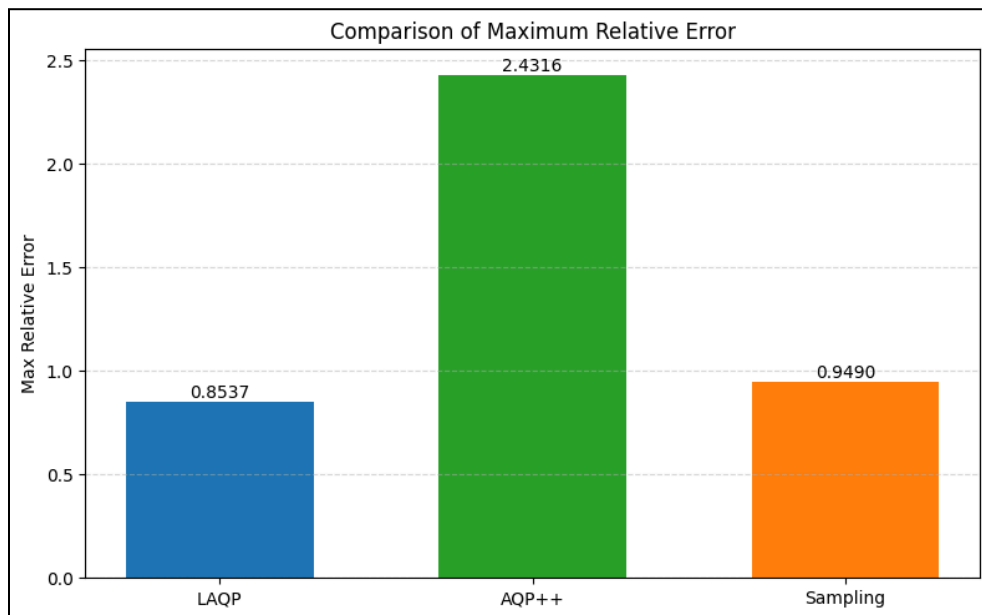


Figure 15: POWER dataset's maximum relative error.

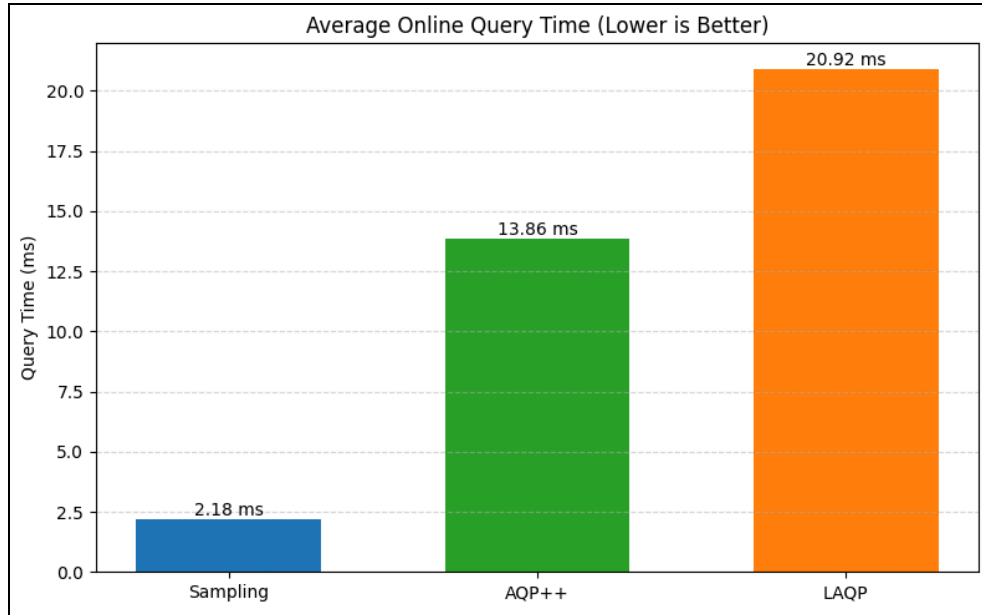


Figure 16: POWER dataset's average query time.

For the Uber dataset, LAQP also achieves the lowest ARE and MSE, confirming its superior performance. Unlike the non-spatial process, LAQP achieves both the minimum relative error and the smallest maximum relative error. LAQP still incurs the highest average query time compared to the other two methods.

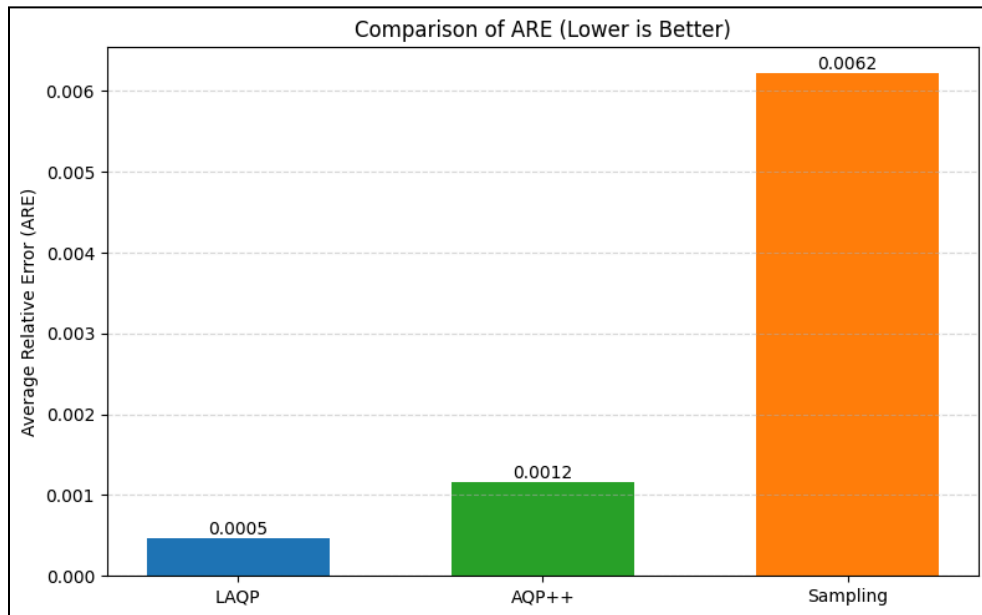


Figure 17: Uber dataset's ARE.

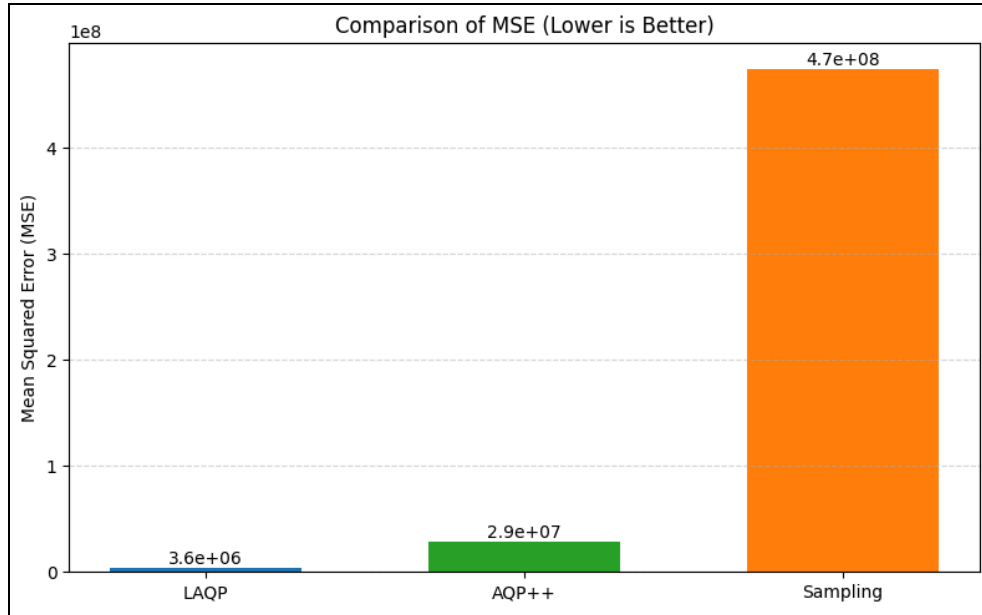


Figure 18: Uber dataset's MSE.

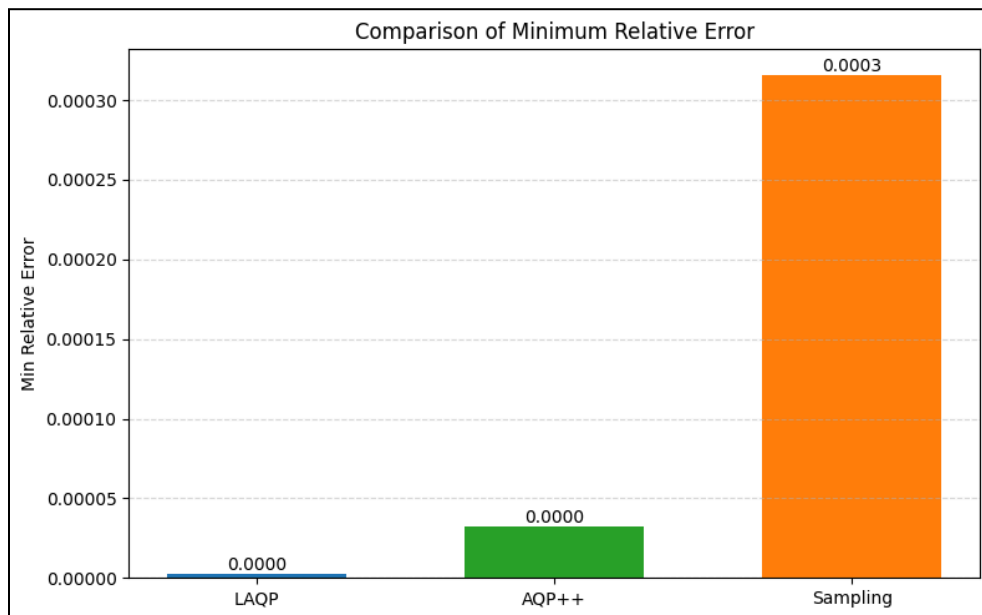


Figure 19: Uber dataset's minimum relative error.

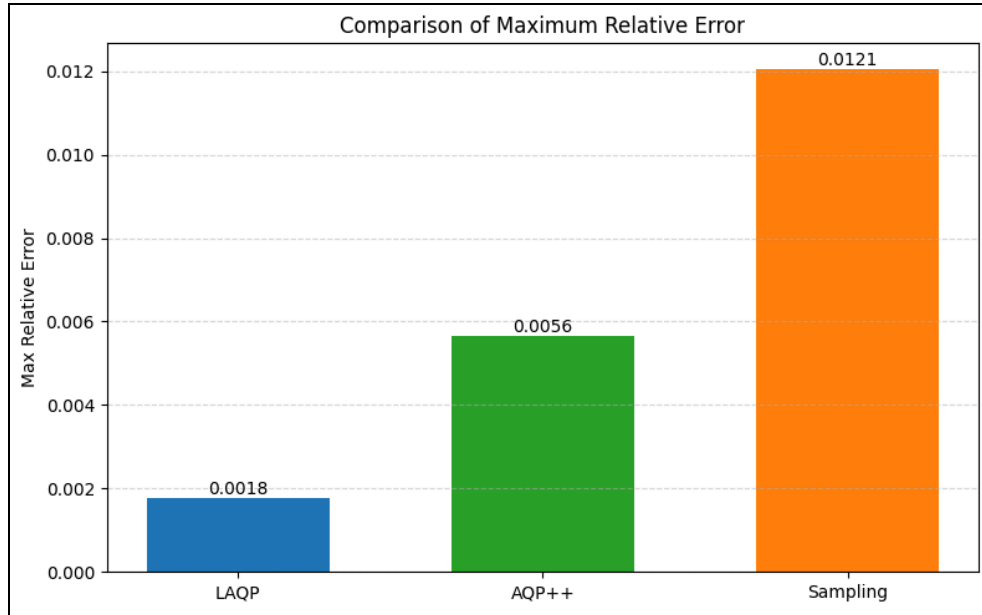


Figure 20: Uber dataset's maximum relative error.

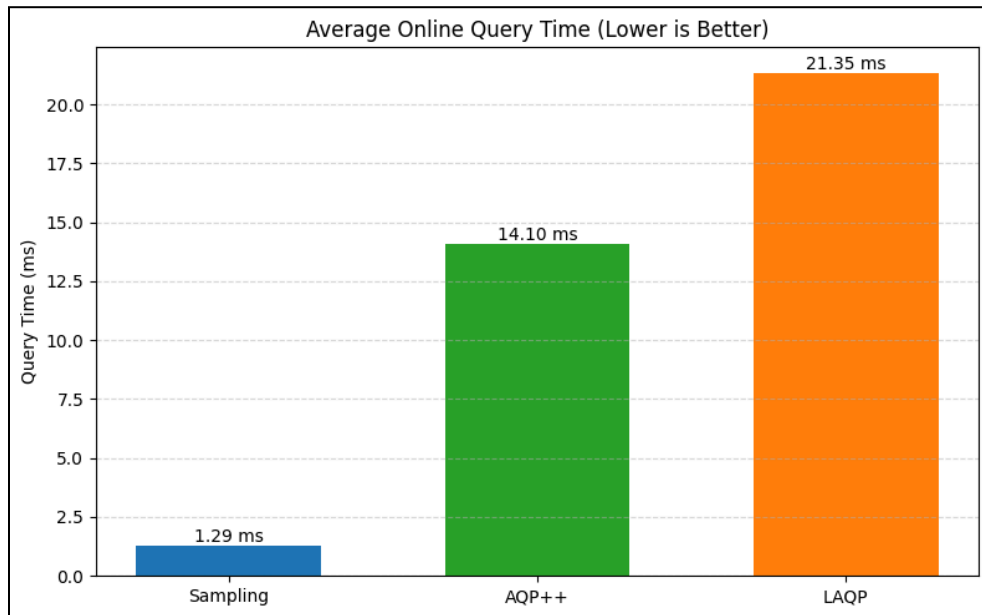


Figure 21: Uber dataset's average query time.

Demo

We developed a demo application using Streamlit, which hosts a website on localhost. Users can select cities such as Brooklyn or Manhattan, specify latitude and longitude bounds, and use a calendar bounded by the Date/Time range in the Uber Pickups dataset. Additionally, users can draw a bounding box (bbox) to customize the query.

LAQP Uber Pickups COUNT Demo

Select City Region

Manhattan

Apply City Region

Start Date

2014/04/01

End Date

2014/09/30

Lat Range (BBox)

40.70 40.82

Lon Range (BBox)

-74.02 -73.93

Generate Random Lat, Lon

Figure 22: Top of the website. Users can select the city region and choose the start/end date, adjust the bbox manually or randomly.



Figure 23: Custom bbox. Users can draw a bbox on the map.

Query BBox Preview (Matching Points Up to 1000)

Query Search

Show Matching Points on Map

Exact COUNT: 3662230

LAQP Approx COUNT: 3661009

Difference: 1220.9859 (Rel Error: 0.0003)

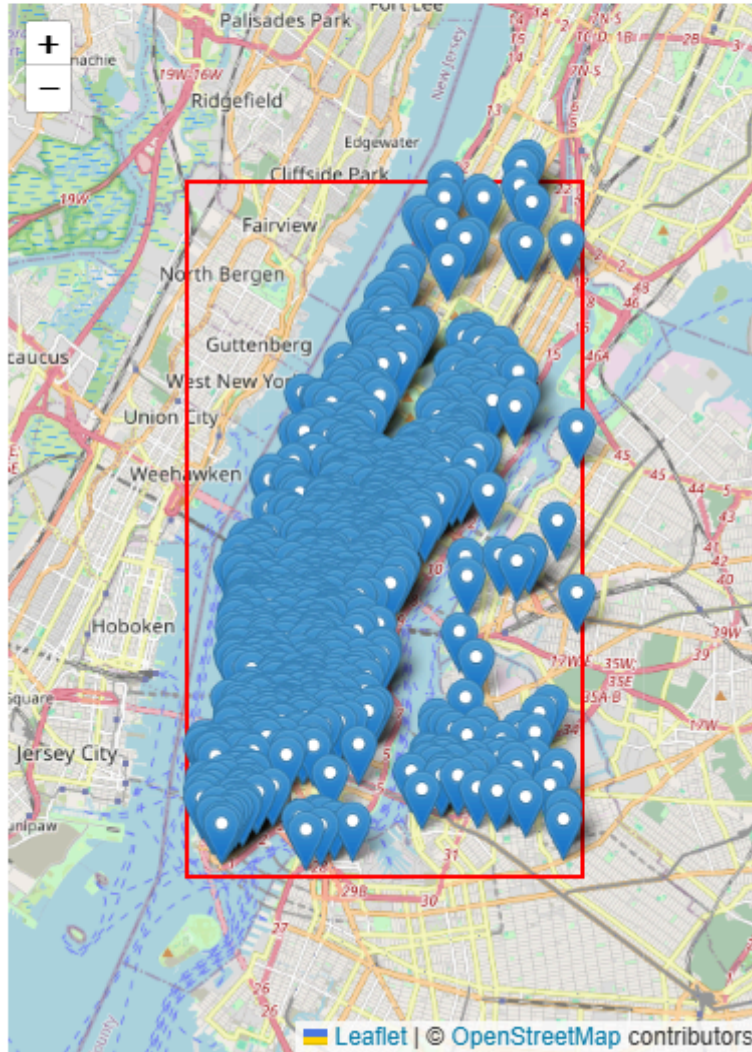


Figure 24: Matching points show on map. After query search, the website will show the LAQP result and difference, the matching points according to the query will also show in the bbox.

6. Conclusion

This project successfully reproduces and implements the LAQP (Learning-based Approximate Query Processing) framework and extends its evaluation to a spatial database context. While the original LAQP method targets general big data querying, this work demonstrates its practical effectiveness on spatial-temporal datasets, particularly for range-based aggregation queries.

The project includes:

- A full implementation of LAQP, including sampling-based estimation and a learned error prediction model.
- Comprehensive benchmarking against two baseline methods: SAQP and AQP++.
- Experimental evaluation on both non-spatial (POWER) and spatial-temporal (NYC Uber Pickups) datasets.
- An **interactive Streamlit-based demo** that enables real-time spatial queries using bounding boxes and temporal filters.

From this project, the following insights are obtained:

- Sampling-based AQP alone is insufficient for spatial datasets with non-uniform distributions.
- Range-based correction (AQP++) improves accuracy but is limited by its reliance on spatial similarity.
- Error-based correction using machine learning (LAQP) provides the **best accuracy**, especially when historical query patterns are available.
- There exists a clear **accuracy–runtime trade-off**, as LAQP incurs additional query processing time due to model inference.

7. Others

	week9	week10	week11	week12	week13	week14	week15	week16	week17	Done
Env setup										V
Reproduce LAQP										V
Spatial adaption										V
Interactive system										V
Experiments And Demo										V

Name/ID	role	workload
汪詩揚/114598020	LAQP, Demo, github, ptt, report	55%
謝豐廷/114598036	AQP++, github, ptt	30%
鄭傳脩/113598022	Github, ppt	15%

Reference

- [1] Peng, J., Zhang, D., Wang, J., Pei, J., 2018. AQP++: connecting approximate query processing with aggregate precomputation for interactive analytics, in: Proceedings of the 2018 International Conference on Management of Data, SIGMOD Conference 2018, Houston, TX, USA, June 10-15, 2018, pp. 1477–1492. URL: <https://doi.org/10.1145/3183713.3183747>, doi:10.1145/3183713.3183747.
- [2] Zhang, Meifan, and Hongzhi Wang. "LAQP: Learning-based approximate query processing." *Information Sciences* 546 (2021): 1113-1134.